INACTIVE FOR NEW DESIGN AS OF JULY 13, 1995.

INCH-POUND

MIL-M-38510/135B
AMENDMENT 2
18 AUGUST 1995
SUPERSEDING
MIL-M-38510/135B
AMENDMENT 1
09 Feb 1995

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, LOW OFFSET OPERATIONAL AMPLIFIERS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/135B, dated 11 December 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2.1, Add the following sentence; "Devices may be monolithic or they may consist of two separate independent die."

* 1.3, Junction temperature, add new footnote 2/ which states "For short term test (in the specific burn-in and steady state life test configuration when required and up to 168 hours maximum), T_J = +175°C."

PAGE 2

3.1, Detail specifications, delete in its entirety and substitute "3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 3

Move page number "3" from the left side of page to the bottom and move "MIL-M-38510/135B" from the right side of the page to the top of the page.
Table I, add new footnote "1/" to the conditions column.
Table I, Input offset voltage tests, footnote "1/" will be renumbered as footnote "2/" (2 times).
Table I, Input offset voltage tests, footnote "2/" will be renumbered as footnote "3/" (2 times).
Table I, Input offset voltage tests, footnote "3/" will be renumbered as footnote "4/" (2 times).
Table I, Input bias current tests, footnote "1/" will be renumbered as footnote "2/" (4 times).
Table I, Input bias current tests, footnote "3/" will be renumbered as footnote "4/" (2 times).
Table I, Input offset current tests, footnote "1/" will be renumbered as footnote "2/" (2 times).

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Table I, add new footnote "1/" to the conditions column.
Table I, Adjustment for input offset tests, footnote "1/" will be renumbered as footnote "2/" (2 times).
Table I, Output short circuit current tests, footnote "4/" will be renumbered as footnote "5/" (2 times).
Table I, Supply current tests, footnote "1/" will be renumbered as footnote "2/" (2 times).
Table I, Supply current tests, footnote "5/" will be renumbered as footnote "6/" (2 times).
Table I, Open loop voltage gain (single ended), A_{VS}, tests, footnote "6/" will be renumbered as footnote "7/" (2 times).

AMSC N/A FSC 5962

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PAGE 5

Table I, add new footnote "1/" to the conditions column.
Footnotes, Add new footnote as follows, "1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified in accordance with the manufacturer's QM plan."
Footnote "1/" will be renumbered as footnote "2/".
Footnote "2/" will be renumbered as footnote "3/".
Footnote "3/" will be renumbered as footnote "4/".
Footnote "4/" will be renumbered as footnote "5/".
Footnote "5/" will be renumbered as footnote "6/".
Footnote "6/" will be renumbered as footnote "7/".

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Table II, note 1, delete "4.2c" and substitute "4.2d".

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Figure 5, device type 03, delete "+18 V ($+V_{CC}$)" and substitute "+15 V ($+V_{CC}$)", delete "-18 V ($-V_{CC}$)" and substitute "-15 V ($-V_{CC}$)".

PAGE 14

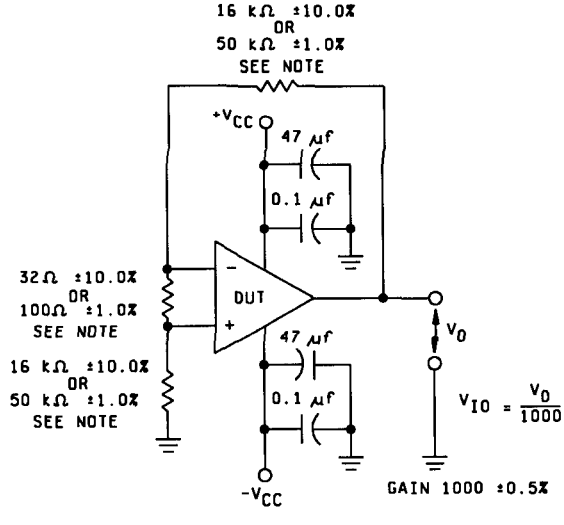
* FIGURE 6, Accelerated burn-in and life test circuit, delete the title and substitute following; "Accelerated burn-in, life test, and class S reverse bias burn-in circuit."

PAGE 15

Figure 7, delete "741" on the nulling amplifier and substitute "741 or equivalent".
Figure 7, Add a capacitor across the input resistors R1 and R2 and reference Note 3.

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* FIGURE 8, voltage offset test circuit, delete and substitute the following;



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* FIGURE 8, voltage offset test circuit, add the following note 4;
"4. Resistors 16 kΩ ±10.0%, 32 Ω ±10%, and 16 kΩ ±10.0% must be used together or resistors 50 kΩ ±1.0%, 100 Ω ±1%, and 50 kΩ ±1.0% must be used together.

PAGE 22

Table III, subgroup 2, test number 15, delete from subgroup 2 and move it to page 24, table III, subgroup 5.

PAGE 23

Table III, subgroup 3, test number 28, delete from subgroup 3 and move it to page 24, table III, subgroup 6.

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* TABLE III, test numbers 5, 6, 20, and 21, under the equation column, delete "77" and substitute "76.9".
Table III, test numbers 5, 6, 20, and 21, equation column, delete "66" and substitute "77" (4 places).
Table III, test numbers 7, equation column, delete "32.25" and substitute "37.04".
Table III, subgroup 2, test number 15, delete from subgroup 2 and move it to page 28, table III, subgroup 5.

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* TABLE III, test numbers 33 and 34, under the equation column, delete "77" and substitute "76.9".
Table III, subgroup 3, test number 28, delete from subgroup 3 and move it to page 28, table III, subgroup 6.
Table III, test numbers 33 and 34, equation column, delete "66" and substitute "77" (2 places).
Table III, test numbers 22 and 35, equation column, delete "32.25" and substitute "37.04"(2 places).

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* TABLE III, test numbers 65, 66, and 67, under measured pin units column, delete "nV" and substitute "pA".
Table III, test numbers 65, 66, and 67, measured pin units column, delete "pA" and substitute "nV".
Footnote 1/, delete footnote 1/ and substitute the following: "1/ All tests apply to figure 7, unless otherwise specified. For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified in accordance with the manufacturer's QM plan."

PAGE 30

4.1, Sampling and inspection, delete in its entirety and substitute "4.1 Sampling and Inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

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Table IV, delete title of table and substitute "TABLE IV. Group C end-point and Group B, class S, electrical parameters. (V_{CM} = 0 V, $\pm V_{CC}$ = ±15 V, for all device types.) T_A = +25°C for Group C end-point limits, $-55 \leq T_A \leq +125°C$ for Group B, class S, end-point limits."

Table IV, $-55 \leq T_A \leq +125°C$ for Group B, class S devices, add values as follows:

V_{IO} test, device types 01, 03, and 05, delete "-100 µV" min and "+100 µV" max and substitute "-135 µV" min and "+135 µV" max.

V_{IO} test, device type 02, delete "-175 µV" min and "+175 µV" max and substitute "-300 µV" min and "+300 µV" max.

V_{IO} test, device type 04, delete "-180 µV" min and "+180 µV" max and substitute "-280 µV" min and "+280 µV" max.

$+I_{IB}$ and $-I_{IB}$ tests, device type 01, delete "-3 nA" min and "+3 nA" maxand substitute "-5 nA" min and "+5 nA" max.

$+I_{IB}$ and $-I_{IB}$ tests, device type 02, delete "-4.5 nA" min and "+4.5 nA" max and substitute "-7.5 nA" min and "+7.5 nA" max.

$+I_{IB}$ and $-I_{IB}$ tests, device types 03, 04, and 05, delete "-50 nA" min and "+50 nA" max"-70 nA" min and "+70 nA" max.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modification, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any in accuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

Civil Agency Coordinating Activity:

NASA - NA

Preparing activity:
DLA - ES

Agent:
DLA - ES

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